Set notation and sample space

1 Heads and Tails

2
a $\{1,2,3,4,5,6\}$ b $\{1,2\}$ c $\{3,6\}$ d $\{2,4,6\}$

3
a $1,2,3,4,5,6$ b $1,2$

4
a $\{2,4,6,8,10,12,14\}$ b $\{-7,-6,-5,-4\}$
c $\{1,3,5,7,9,11\}$ d $\left\{1,\frac{1}{10},\frac{1}{100},\frac{1}{1000},\frac{1}{10\,000},\frac{1}{100\,000}\right\}$
e $\{2,4,8,16,32,64\}$

5
a $\{1,2,4,8,16\}$ b $\{1,2,4,8\}$

Set notation (Extended)

6 $\{x : x \text{ is an even number and } 2 \leq x \leq 8\}$

7
a True b True c False d True

8
a $\subseteq$ b $\subseteq$ c $\nsubseteq$ d $\nsubseteq$

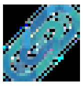
9
a $\in$ b $\notin$

10
a $\{\text{g, a, l, o, p}\}$ b Yes
c $\{\text{p, a, r, l, e, o, g, m}\}$

11 The set of all the elements in the universal set that are not in set A.

12

a $A = \{0, 1, 2, 3, 4, 5\}$



b Yes

13

a 11 b 12 c 7 d 16

14

a No b Yes c No d Yes

15

a Because there is only one point that satisfies both equations, and it occurs at the point of intersection of the lines.

b $(10, 26)$

Operations on sets

16

a The set containing all the elements that are members of set A or of set B or of both sets.

b The set containing all the elements that are common to both set A and set B.

17

a Yes

b $\{\text{cauliflowers}\}$

18 $\{3, 7\}$

19

a $\{0, 4, 16, 36, 64\}$

b $\{5, 10, 15, 20, 25, 30, 35, 40, 45\}$

20

a $\{1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36\}$

b $\{1, 2, 3, 4, 6, 12\}$

21

a $\{1, 2, 3, 4, 5, 6\}$ b $\{1, 2, 3, 4, 5, 6, 7, 8\}$ c $\{1, 2, 3, 4, 5, 6\}$ d $\{1, 2, 3, 4, 5, 6, 7, 8\}$

Complement of sets (Extended)

22

a

i $\{\text{roses, daisies}\}$

ii $\{\text{lillies}\}$

iii $\{\text{roses, daisies, lillies, sunflowers}\}$

iv $\{\text{roses, daisies}\}$

b Yes

23

a Yes

b 5

c $\{7, 13, 12, 26\}$

d $U = \{7, 8, 9, 12, 13, 21, 26, 28, 30\}$

e The empty set $\emptyset$.

24

a $\{5, 6, 7\}$

b $\emptyset$

25

a People who do not like softball.

b People who like either football, softball, or swimming.

26

a Yes

b No

c $\{-5, 3\}$

d $\{11\}$

e 19

f 16

Venn diagrams

27

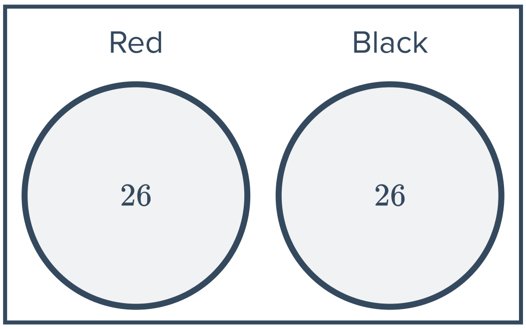
a Venn diagram 4

b Venn diagram 3

c Venn diagram 2

d Venn diagram 1

28



29

a $A \cap B = \{7, 8\}$

b $A \cup B = \{2, 4, 6, 7, 8, 10, 14\}$

Venn diagrams (Extended)

30

a $A' = \{7, 14, 12, 10\}$

b $B' = \{4, 5, 1, 12, 10\}$

31

a $A = \{1, 3, 4, 5, 9, 10\}$

b $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

c $B' = \{1, 2, 9, 10, 11, 13\}$

32

a $A \cap B' = \{3, 4, 6\}$

b $(A \cup B)' = \{15, 60\}$

33

a Yes

b Yes

34

a $\{4, 5, 9\}$

b $\{1, 2, 3, 7, 9, 10, 11, 12, 13, 14\}$

c $\{4, 5\}$

d $\{3, 4, 5, 9\}$

e $\{1, 2, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

35

a Yes

b No

36

a 2

b 5

c 3

d 4

e 8

f 7

g 1

h 6

37

a

	Play Rugby League	Don't play Rugby League
Play Rugby Union	43	185
Don't play Rugby Union	179	146

b

i 179

ii 222

iii 228

iv 185

v 331

vi 325